Input system:
Dependent variables:y_2, y_1
Independent variables:x_2, x_1
Functions:

$$y_1 = x_1^5 \sin(x_2)$$

$$y_2 = \ln(x_1)^3 \cos(x_2)$$

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Result:
Dependent variables:$y_2, s_4, s_2, s_3, s_1, y_1$
Independent variables:x_2, x_1
Additional variables:

$$s_1 = \sin(x_2)$$

$$s_2 = \cos(x_2)$$

$$s_3 = \ln(x_1)$$

$$s_4 = \text{inv}(x_1)$$

Functions:

$$y_1 = x_1^5 s_1$$

$$y_2 = s_3^3 s_2$$

Equations:

$$\frac{\partial s_1}{\partial x_2} = s_2$$

$$\frac{\partial s_1}{\partial x_1} = 0$$

$$\frac{\partial s_2}{\partial x_2} = (-1) s_1$$

$$\frac{\partial s_2}{\partial x_1} = 0$$

$$\frac{\partial s_3}{\partial x_2} = 0$$

$$\frac{\partial s_3}{\partial x_1} = s_4$$

$$\frac{\partial s_4}{\partial x_2} = 0$$

$$\frac{\partial s_4}{\partial x_1} = (-1) s_4^2$$